PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

KILN TAIL POST 窑尾岗位

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Table of Contents

DOCUMENT CHANGE STATUS 2

Table of Contents 3

1. Purpose 目的 4
2. Functional Description 功能说明 4
3. References 参考 4
4. PPE 个人保护装备 5
5. Hazards & Control 危害与控制 5
6. Equipment 工具/ 设备 6
7. Procedure 程序 6
8. Kiln Tail Post 窑尾岗位 Questionnaire 23
9. Appendix 24

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the kiln tail post under complete closure.

此程序涵盖操作员对完全关闭下的窑尾岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the kiln tail post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the kiln tail post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对窑尾岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定窑尾岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.
  MTI-CP-SOP岗位操作规程
- High Angle Belt Conveyor(4303-BCV-007).
  高倾角皮带输送机(4303-BCV-007)
- Belt Conveyors(4303-BCV-008).
  皮带输送机(4303-BCV-008)
- Bucket Elevator(4303-BEL-001).
  斗式提升机(4303-BEL-001)
- Kiln Tail Cover(4304-KTC-001).
  窑尾罩(4304-KTC-001)
- Kiln Tail Cooling Fan(4304-BLO-003/004)
  窑尾冷却风机(4304-BLO-003/004)
- Settling Chamber(4313-SEC-001).
  重力沉降室(4313-SEC-001)
- Buried Scraper Conveyor(4313-SCV-001/002/003).
  埋刮板输送机(4313-SCV-001/002/003)
- Screwed Conveyor(4304/4313-PCC-001).
  管链输送机(4304/4313-PCC-001)
- Fan(4313-FAN-001/002/003/004/005/006).
  风机(4313-FAN-001/002/003/004/005/006)

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4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服



防毒口罩　防尘口罩

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Empyrosis; 7. Object falling from high altitudes; 8. Poisoning; 9. Noise. 1. 机械伤害; 2. 灰尘; 3. 触电; 4. 湿滑的条件，绊倒危险; 5. 窒息; 6. 烫伤; 7. 高空坠物; 8. 中毒; 9. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear good labor protection goods; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Wear antigas masks as required; 9. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩; 2. 按要求佩戴防尘口罩; 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效; 4. 上下楼梯请扶好扶手; 5. 清仓或维修作业执行审批程序; 6. 穿戴好劳保护品; 7. 穿戴好劳保护品，检查各操作平台的

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
	物品堆放安全性； 8. 按要求佩戴防毒口罩； 9. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4303-BIN-001	Accident Bunker 事故料仓
4313-TNK-001	Soot Can 烟尘罐
4313-ADG-001 ~ 008	Star Ash Exhaust Valve 星型排灰阀
4303-ADG-029,4304-ADG-001	High Temperature Resistant Double-Layer Flap Valve 耐高温双层翻板阀
4303-ROD-001	Bar Valve 棒条阀
4303-CHE-012/013	In and Out of The Material Slip Groove 进出料溜槽
4304-CHE-001	Kiln Tail Groove 窑尾溜槽
	Clean the site tools 清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 4. 与前后工序取得联系，并发出明确指令。
2	Inspect High Angle Belt Conveyor (4303-BCV-007) 检查高倾角皮带输送机(4303-BCV-007)	1. Good bearing lubrication and sealing, proper oil level, upper and lower rolling lubrication, flexible operation, 2. Whether there are materials between the first and tail wheels and the tape, and whether the cleaning device of the first round is in good condition;

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№	Stage	Key Points
		3. Comprehensively check whether all parts of the belt conveyor are in good condition, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 4. Whether the electrical facilities and control system are in good condition, whether the protective device is firm, and whether the leakage baffle and lining plate are damaged; 5. Check whether the buffer olroller under the impact load is offset; 6. Whether the installed electrical facilities and signal devices are in good and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, watch whether the tape running direction is correct, whether the belt is off; whether the belt skirt falls off, whether the belt edge is off, tear phenomenon; 8. After checking the start button and contacting the post, press the start button; 9. Check whether there is any skid phenomenon in the process of the belt upward; 10. During operation, check whether the emergency stop device of the belt conveyor, belt correction and speed sensor are sensitive. 1. 各部轴承润滑与密封良好，减速机油位适当，上下托滚润滑良好，运转灵活， 2. 首、尾轮磨损情况与胶带间是否夹有物料，首轮的清扫装置是否完好； 3. 全面检查皮带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 4. 电器设施和控制系统是否完好，防护装置是否牢固，漏料挡板和衬板是否损坏； 5. 检查承受冲击载荷部位的缓冲托辊是否偏移； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制箱置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏；皮带裙边有无稳钉脱落，皮带边沿有无脱线、撕裂现象； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 检查皮带上行过程中有无打滑现象；

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№	Stage	Key Points
		10. 运转过程中，检查高倾角皮带输送机急停装置、皮带纠偏和测速传感器是否控制灵敏
3	Inspect Belt Conveyors (4303-BCV-008) 检查皮带输送机 (4303-BCV-008)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查皮带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏；

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№	Stage	Key Points
		4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查皮带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查皮带输送机急停装置、皮带纠偏和测速传感器是否控制灵敏
4	Inspect Bucket Elevator (4303-BEL-001) 检查斗式提升机(4303-BEL-001)	1. Check whether the entrance and outlet are blocked; 2. Check whether the reducer has no leakage and lubrication, and whether the position of the lubricating oil is between the upper and lower limits of the oil standard; 3. Check whether the head and tail sprocket bearings and motor bearings are well lubricated; 4. Check the hydraulic coupling for no leakage and lubrication; 5. Check whether the reverse check device is normal; 6. Check whether the protective devices and safety facilities are in good condition, complete and firm, and all access doors and observation doors are closed; 7. Check whether the chain plate, shaft pin and hopper are in good condition, and whether the hopper bolt is loose; 8. Check whether the operating parts are flexible, whether the connecting parts and bolts are fastened, whether the motor and electrical control equipment are safe and reliable, and determine whether the rotation direction of the motor is correct; 9. Check the first start for impact, violent shaking or abnormal noise before the start after stopping for a long time; 10. The tail wheel of the bucket lift is allowed to swing slightly when running. If the unilateral move is large, the tightness of the bearing stud should be adjusted appropriately.

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№	Stage	Key Points
		1. 检查入口和出口是否堵塞； 2. 检查减速器有无泄漏和润滑，润滑油的位置是否在油标准的上下限之间； 3. 检查头尾链轮轴承和电机轴承是否润滑良好； 4. 检查斗提机尾部料斗是否有掩埋、转动灵活； 5. 检查逆止器装置是否正常； 6. 检查防护装置和安全设施是否完好完整、齐全牢固，所有通道门和观察门关闭； 7. 检查链板、轴销、料斗是否完好，料斗螺栓是否松动； 8. 检查操作部件是否灵活，连接部件和螺栓是否紧固，电机和电气控制设备是否安全可靠，判明电机旋转方向是否正确； 9. 第一次启动和在长时间停止后再启动前，检查链斗是否有冲击、剧烈晃动或异常噪音； 10. 斗式提升机尾轮运行时允许有上下轻微摆动，如果单边摆动较大适当调整轴承部位螺柱的松紧度
5	Inspect Kiln Tail Cover (4304-KTC-001) 检查窑尾 (4304-KTC-001)	1. Check that the kiln tail observation door (cover) is intact; 2. Check whether the instrument of the kiln tail cover is in good condition; 3. Whether the cooling air device is in good condition, and the bolts are not loose; 4. Check whether the sealing of the kiln tail release valve is good; 5. Check whether the refractory material of the kiln tail has fallen off; 6. Check whether the scales of the kiln tail cover are close to the rotary kiln cylinder, whether the scales sealing device is adjustable and normal, and whether the wire rope of the scales is effective. 1. 检查窑尾观察门(盖)完好； 2. 检查窑尾罩的仪表是否完好； 3. 冷却风装置是否保持良好状态，各处螺栓无松动； 4. 检查窑尾放散阀密封是否良好； 5. 检查窑尾耐火材料是否有脱落； 6. 检查窑尾罩鱼鳞片是否紧贴回转窑筒体，鱼鳞片密封装置是否可调、是否正常，鱼鳞片紧固钢丝绳是否有效
6	Inspect Kiln Tail Cooling Fan (4304-BLO-003/004), Fan	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena;

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№	Stage	Key Points
	(4313-FAN-001/002/003/004/005/006) 检查窑尾冷却风机 (4304-BLO-003/004)，风机 (4313-FAN-001/002/003/004/005/006)	2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller) 1. 手动盘车2～3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)
7	Inspect Settling Chamber (4313-SEC-001) 检查重力沉降室 (4313-SEC-001)	1. Check whether the population is normal, observe whether there are problems in the windshield wall and the inner next door through the population, and check whether the outer wall is normal; 2. Check the ventilation pipe of the outer wall, whether the hot air inlet and outlet pipe in the inner door, the cold air in the inner door, the hot air inlet and outlet pipe in the outer wall, and the cold air inlet pipe in the outer door are intact; 3. Check whether the wind temperature and pressure before and after the cold air of the inner door and outer walls is normal; 4. Check whether the 14 thermocouple devices are in good condition; 5. Check whether the height and gray level measurement of the ash bucket shows that the instrument is in good condition. 1. 检查人孔是否正常，通过人孔观察挡风墙、内隔壁有无问题，检查外墙体是否正常；

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№	Stage	Key Points
		2. 检查外墙体通风管，内隔壁热风输进出管，内隔壁冷风输进出管，外围壁热风进出管，外隔壁冷风输进出管是否完好； 3. 检查内隔壁和外围壁冷风进出前后的温度、风压是否正常； 4. 检查14个热电偶装置是否完好； 5. 检查集灰斗高低灰位测量显示仪表是否完好
8	Inspect Buried Scraper Conveyor (4313-SCV-001/002/003) 检查埋刮板输送机 (4313-SCV-001/002/003)	1. Check whether the bolts are fastened; 2. All bearings, transmission parts and speed reducer shall be equipped with sufficient lubricating oil; 3. Comprehensively check whether all parts of the conveyor are intact and whether the tightness of the scraper chain is appropriate; 4. Before driving, it should be determined whether the rotation direction of the motor meets the work requirements; 5. Check whether the scraper chain and drive device are in good condition; 6. Manual drive and observe whether the scraper chain touches or deviates with the housing card. 1. 检查各螺栓是否紧固； 2. 所有轴承、传动部件和减速器内应有足够的润滑油； 3. 全面检查输送机各部分是否完好无损，刮板链条松紧度是否合适； 4. 开车前应判明电机旋转方向是否符合工作要求； 5. 检查刮板链条和驱动装置是否完好无损； 6. 手动盘车，观察刮板链条是否与壳体卡碰或跑偏
9	Screwed Conveyor (4304/4313-PCC-001) 检查管链输送机 (4304/4313-PCC-001)	1. Check whether the bolt connection of the groove bottom plate and the chain plate is loose; 2. Check whether the tightness of the traction chain is normal; 3. Check whether the chain plate conveyor should inject lubricant in time at each lubrication point is normal and whether oil leakage; 4. Check whether the transmission parts are falling off, are not firmly tightened, or have abnormal friction;

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№	Stage	Key Points
		5. Determine whether the rotation direction of the motor meets the work requirements; 6. Check whether the conveying roll is partial. 1. 检查槽底板及链板的螺栓连接是否有松动现象； 2. 检查牵引链的松紧度是否正常； 3. 检查链板输送机在各润滑点应及时注入润滑剂是否正常，是否漏油； 4. 检查传动部件是否有脱落、拧紧不牢或摩擦异常； 5. 判明电机旋转方向是否符合工作要求； 6. 检查输送时卷料是否偏料
10	Check the other Equipment and Facilities 检查其它设备设施	1. Accident Bunker(4303-BIN-001) a. Check whether the inlet and outlet pipes and auxiliary pipes and flanges of each warehouse meet the requirements; b. Check the material level control assembly for normal operation. 1. 事故料仓(4303-BIN-001) a. 检查料仓出入口管路及附属管路、法兰是否符合要求； b. 检查料位控制组件是否正常。 2. Soot Can(4313-TNK-001) a. Check whether the inlet and outlet pipes and auxiliary pipes and flanges of each warehouse meet the requirements, and whether the seal of each warehouse is in good condition; b. Check the material level control assembly for normal operation; c. Check whether the inlet chute of the smoke tank is normal. 2. 烟尘罐(4313-TNK-001) a. 检查烟尘罐出入口管路及附属管路、法兰是否符合要求，密封是否完好； b. 检查料位控制组件是否正常； c. 检查烟尘罐入口溜槽否正常。 3. The star type ash valve of gravity settling chamber(4313-ADG-001 ~ 008)

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№	Stage	Key Points
		a. Check that the rotor is rotated in the correct direction. 3. 重力沉降室集灰斗的星型排灰阀(4313-ADG-001 ~ 008) a. 检查转子的旋转方向是否正确。 4. Bar Valve(4303-ROD-001) a. Whether the frame installation direction is correct, whether the connecting bolts between the frame and the bin are fastened, and whether the material leaks between the frame and the bin; b. Whether the spot check of the rod bar is flexible; c. Whether the bar valve is closed. 4. 棒条阀(4303-ROD-001) a. 框架安装方向是否正确，框架与料仓之间的连接螺栓是否紧固，框架与料仓之间是否漏料； b. 棒条的抽查是否灵活； c. 棒条阀是否处于关闭状态。 5. High temperature resistant double-layer flap valve(4303-ADG-029,4304-ADG-001) a. Turn on the power supply and check whether the motor steering direction is consistent with the arrow mark; b. Check whether the electric double-layer ash discharge valve weight block stops and returns, whether the swing rod is loose and falls off, and whether the flange connection bolt is loose and falls off; c. Fill the bearing base with lubricating grease (general lithium grease, 30g /). 5. 耐高温双层翻板阀(4303-ADG-029,4304-ADG-001) a. 接通电源，检查电机转向与箭头标示方向是否一致； b. 检查电动双层卸灰阀配重块止退销、摆杆是否松动脱落，法兰连接螺栓是否有松动脱落；

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№	Stage	Key Points
		c. 轴承座加注润滑油脂(通用锂基脂,30g/处)。 6. Kiln-tail chute (4304-CHE-001), Bucket elevator (4303-BEL-001) feed chute (4303-CHE-012) and discharge chute (4303-CHE-013), Belt Conveyors (4303-BCV-008) drain the groove, Gravity settling chamber collecting ash bucket. a. Check whether the feeding chute is in good condition; b. Check whether the blocking material detection switch of the lower feeding chute is normal. 6. 窑尾溜槽(4304-CHE-001),斗式提升机(4303-BEL-001)进料溜槽(4303-CHE-012)和出料溜槽(4303-CHE-013),皮带输送机(4303-BCV-008)下料溜槽，重力沉降室集灰斗 a. 检查各下料溜槽/斗是否完好； b. 检查下料溜槽/斗的堵料检测开关是否正常。 7. Valve a. Check whether the soot tank feeding outlet valve (200 KG) is normal b. Buried scraper conveyor (4313-SCV-001/002) feeding star discharge valve (400SD) is normal; c. Kiln tail cooling fan(4304-BLO-003/004)Whether the inlet and outlet butterfly valve (500BF) is normal; d. Fan (4313-FAN-001/002/003/004/005/006) outlet butterfly valve (400BF) is normal; e. Whether the kiln tail release valve (***) is normal; f. Check whether the outer wall ventilation pipe valve (***), the next hot air inlet and outlet pipe valve (***, ***), the next cold air inlet and outlet pipe valve (***, ***), the outer wall hot air inlet and outlet pipe valve (***, ***) and the next cold air inlet and outlet pipe valve (***, ***) are flexible. 7. 阀门 a. 检查烟尘罐下料口阀门(200KG)是否正常； b. 埋刮板输送机(4313-SCV-001/002)入料星型卸料阀(400SD)是否正常；

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№	Stage	Key Points
		c. 窑尾冷却风机(4304-BLO-003/004)进出蝶阀(500BF)是否正常； d. 风机(4313-FAN-001/002/003/004/005/006)出口蝶阀(400BF)是否正常； e. 窑尾放散阀(***)是否正常， f. 检查外墙体通风管阀门(***)、内隔壁热风输进出管阀门(***,***)、内隔壁冷风输进出管阀门(***,***)、外围壁热风进出管阀门(***,***)和外隔壁冷风输进出管阀门(***,***)是否灵活自如
11	Inspect Materials Level 检查物料料位	1. Check whether the material is stored on the high inclination belt conveyor (4303-BCV-007) and belt conveyor (4303-BCV-008), and there is no adhesive deposit on the surface of the head and tail wheels and the joint of the belt; 2. Check the bottom of the bucket elevator (4303-BEL-001); 3. Check whether the kiln tail chute (4304-CHE-001) of the kiln tail cover (4304-KTC-001),Bucket elevator(4303-BEL-001)feed chute (4303-CHE-012) and discharge chute (4303-CHE-013) has any materials; 4. Check whether the ash bucket in the gravity settlement chamber (4313-SEC-001) has ash; 5. Check whether the material is stored in the buried scraper conveyor (4313-SCV-001/002/003) and the pipe chain conveyor (4304-PCC-001,4313-PCC-001); 6. Check whether the dust tank (4313-TNK-001) has ash. 1. 检查高倾角皮带输送机(4303-BCV-007)、皮带输送机(4303-BCV-008)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在； 2. 检查斗式提升机(4303-BEL-001)底部有无物料； 3. 检查窑尾罩(4304-KTC-001)的窑尾溜槽(4304-CHE-001),斗式提升机(4303-BEL-001)进料溜槽(4303-CHE-012)和出料溜槽(4303-CHE-013)有无物料； 4. 检查重力沉降室(4313-SEC-001)集灰斗是否存

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№	Stage	Key Points
		灰； 5. 检查埋刮板输送机(4313-SCV-001/002/003)、管链输送机 (4304/4313-PCC-001) 内是否存料； 6. 检查烟尘罐 (4313-TNK-001) 是否存灰
12	Start-Up Of Kiln Tail Post 窑尾岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later; At the same time, Confirm that the process behind the kiln tail post has been started and has the feeding conditions: 3. Normal Start-Up Open screwed conveyor(4313-PCC-001)⟹ Open lower port valve of soot can(200KG)⟹ Open buried scraper conveyor(4313-SCV-003)⟹Open buried scraper Conveyor(4313-SCV-001/002)⟹Open star discharger(400SD)⟹Open star discharger(4313-ADG-001 ~ 008)⟹Open the outlet butterfly valve (400BF), Open fan(4313-FAN-001/002/003/004/005/006)⟹Open the outer wall ventilation pipe valve (***), the next hot air inlet and outlet pipe valve (***, ***), the next cold air inlet and outlet pipe valve (***, ***), the outer wall hot air inlet and outlet pipe valve (***, ***) and the next cold air inlet and outlet pipe valve (***, ***)⟹Open screwed conveyor(4304-PCC-001)⟹Open high temperature resistant double-layer flap valve(4304-ADG-001)⟹Open the access butterfly valve (500BF), Open kiln tail cooling fan(4304-BLO-003/004)⟹Open high temperature resistant double-layer flap valve(4303-ADG-029)⟹Open high angle belt conveyor(4303-BCV-007). 4. After the rotary kiln accident Start-Up Open screwed conveyor(4313-PCC-001)⟹ Open lower port valve of soot can(200KG)⟹ Open buried scraper conveyor(4313-SCV-003)⟹Open buried scraper Conveyor(4313-SCV-001/002)⟹Open star discharger(400SD)⟹Open star discharger(4313-ADG-001 ~ 008)⟹Open the outlet butterfly valve (400BF), Open fan(4313-FAN-001/002/003/004/005/006)⟹Open the outer wall ventilation pipe valve (***), the next hot air inlet and outlet pipe valve

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№	Stage	Key Points
		(***, ***), the next cold air inlet and outlet pipe valve (***, ***), the outer wall hot air inlet and outlet pipe valve (***, ***) and the next cold air inlet and outlet pipe valve (***, ***)⟹Open screwed conveyor(4304-PCC-001)⟹Open high temperature resistant double-layer flap valve(4304-ADG-001)⟹Open the access butterfly valve (500BF), Open kiln tail cooling fan(4304-BLO-003/004)⟹Open High temperature resistant double-layer flap valve(4303-ADG-029)⟹Open bucket elevator(4303-BEL-001)⟹The bucket elevator(4303-BEL-001) is running for 5 minutes, Open bar valve(4303-ROD-001)⟹Open belt conveyors(4303-BCV-008)⟹Open high angle belt conveyor(4303-BCV-007) 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认窑尾岗位后面的工序已经开车并具备给料条件： 3. 正常开机 开启管链输送机(4313-PCC-001)⟹打开烟尘罐下料口阀门(200KG)⟹开启埋刮板输送机(4313-SCV-0003)⟹开启埋刮板输送机(4313-SCV-001/002)⟹打开星型卸料阀(400SD)⟹打开星型排灰阀(4313-ADG-001～008)⟹打开出口蝶阀(400BF)，启动风机(4313-FAN-001/002/003/004/005/006)⟹打开外墙体通风管阀门(***)、内隔壁热风输进出管阀门(***,***)、内隔壁冷风输进出管阀门(***,***)、外围壁热风进出管阀门(***,***)和外隔壁冷风输进出管阀门(***,***)⟹开启管链输送机(4304-PCC-001)⟹打开耐高温双层翻板阀(4304-ADG-001)⟹打开进出蝶阀(500BF)，开启窑尾冷却风机(4304-BLO-003/004)⟹打开耐高温双层翻板阀(4303-ADG-029)⟹开启高倾角皮带输送机(4303-BCV-007)。 4. 回转窑事故后开机

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№	Stage	Key Points
		开启管链输送机(4313-PCC-001)⟹打开烟尘罐下料口阀门(200KG)⟹开启埋刮板输送机(4313-SCV-003)⟹开启埋刮板输送机(4313-SCV-001/002)⟹打开星型卸料阀(400SD)⟹打开星型排灰阀(4313-ADG-001～008)⟹打开出口蝶阀(400BF)，启动风机(4313-FAN-001/002/003/004/005/006)⟹打开外墙体通风管阀门(***)、内隔壁热风输进出管阀门(***,***)、内隔壁冷风输进出管阀门(***,***)、外围壁热风进出管阀门(***,***)和外隔壁冷风输进出管阀门(***,***)⟹开启管链输送机(4304-PCC-001)⟹打开耐高温双层翻板阀(4304-ADG-001)⟹打开进出蝶阀(500BF)，开启窑尾冷却风机(4304-BLO-003/004)⟹打开耐高温电动双层阀(4303-ADG-029)⟹开启斗式提升机(4303-BEL-001)⟹斗式提升机(4303-BEL-001)开机运行5分钟后，打开棒条阀(4303-ROD-001)⟹开启皮带输送机(4303-BCV-008)⟹开启高倾角皮带输送机(4303-BCV-007)
13	The Kiln Tail Post Stops 窑尾岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop: Stop high angle belt conveyor(4303-BCV-007)⟹Stop high temperature resistant double-layer flap valve(4303-ADG-029)⟹Stop kiln tail cooling fan(4304-BLO-003/004), Close the access butterfly valve (500BF)⟹Stop high temperature resistant double-layer flap valve(4304-ADG-001)⟹Stop screwed conveyor(4304-PCC-001)⟹Close the outer wall ventilation pipe valve (***), the next hot air inlet and outlet pipe valve (***, ***), the next cold air inlet and outlet pipe valve (***, ***), the outer wall hot air inlet and outlet pipe valve (***, ***) and the next cold air inlet and

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№	Stage	Key Points
		outlet pipe valve (***, ***)⟹Stop fan(4313-FAN-001/002/003/004/005/006), Close the outlet butterfly valve (400BF)⟹ Stop star discharger(4313-ADG-001 ~ 008) ⟹Stop star discharger(400SD)⟹ Stop buried scraper Conveyor(4313-SCV-001/002)⟹Stop buried scraper conveyor(4313-SCV-003)⟹Stop lower port valve of soot can(200KG)⟹Stop screwed conveyor(4313-PCC-001)⟹ Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序和物料排净顺序停车，各设备都需要延时停止： 停止高倾角皮带输送机(4303-BCV-007) ⟹关闭耐高温电动双层阀(4303-ADG-029)⟹停止窑尾冷却风机(4304-BLO-003/004)，关闭进出蝶阀(500BF)⟹关闭耐高温双层翻板阀(4304-ADG-001)⟹停止管链输送机(4304-PCC-001)⟹关闭外墙体通风管阀门(***)、内隔壁热风输进出管阀门(***,***)、内隔壁冷风输进出管阀门(***,***)、外围壁热风进出管阀门(***,***)和外隔壁冷风输进出管阀门(***,***)⟹停止风机(4313-FAN-001/002/003/004/005/006)，关闭出口蝶阀(400BF)⟹停止星型排灰阀(4313-ADG-001 ~ 008)⟹停止星型卸料阀(400SD) ⟹停止埋刮板输送机(4313-SCV-001/002)⟹停止埋刮板输送机(4313-SCV-003)⟹关闭烟尘罐下料口阀门(200KG)⟹停止管链输送机(4313-PCC-001)⟹ ⑯打扫现场卫生。

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№	Stage	Key Points
		2. Emergency Stop a. When an emergency occurs, Close high angle belt conveyor (4303-BCV-007), fan (4313-FAN-001/002/003/004/005/006), buried scraper conveyor (4313-SCV-001/002), buried scraper conveyor (4313-SCV-003), screwed conveyor (4304/4313-PCC-001), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close high temperature resistant double-layer flap valve (4303-ADG-029), high temperature resistant double-layer flap valve (4304-ADG-001), star discharger (4313-ADG-001~008), kiln tail cooling fan(4304-BLO-003/004) star discharger (400SD), lower port valve of soot can (200KG), the access butterfly valve (500BF), Fan (4313-FAN-001/002/003/004/005/006) outlet butterfly valve (400BF),Close the outer wall ventilation pipe valve (***), the next hot air inlet and outlet pipe valve (***, ***), the next cold air inlet and outlet pipe valve (***, ***), the outer wall hot air inlet and outlet pipe valve (***, ***) and the next cold air inlet and outlet pipe valve (***, ***); at the same time, the upper level of equipment all need to stop; The cooling fan (4304-BLO-003/004) does not stop during the various fault shutdown kiln and rotary kiln sealing operation; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭高倾角皮带输送机(4303-BCV-007)、风机(4313-FAN-001/002/003/004/005/006)、埋刮板输送机(4313-SCV-001/002)、埋刮板输送机(4313-SCV-003)、管链输送机(4304/4313-PCC-001)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭耐高温

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№	Stage	Key Points
		电动双层阀(4303-ADG-029)、耐高温双层翻板阀(4304-ADG-001)、星型排灰阀(4313-ADG-001～008)、星型卸料阀(400SD)、烟尘罐下料口阀门(200KG)、 冷却风机(4304-BLO-003/004)进出蝶阀(500BF)、风机(4313-FAN-001/002/003/004/005/006)出口蝶阀(400BF)，关闭外墙体通风管阀门(***)、内隔壁热风输进出管阀门(***,***)、内隔壁冷风输进出管阀门(***,***)、外围壁热风进出管阀门(***,***)和外隔壁冷风输进出管阀门(***,***)；同时，上一级设备均需停车；各种故障停窑及回转窑保温操作时，窑尾密封机构的冷却风机(4304-BLO-003/004)不停机； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作
14	Permits ＆Isolation Sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Kiln Tail Post 窑尾岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Kiln tail posts in the chloride plant
CP 窑尾岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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